

## A calibration method for handling the temporal drift of solid state gas-sensors

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### Abstract

The reproducibility of chemical sensors is an issue that has to be handled in order to apply sensor-array based instruments for quality control purposes on a routine basis in the laboratory. So far, this problem can only be handled by applying mathematical algorithms to describe the temporal variation of the sensor signal. Sensor drift in a commercial solid state based sensor array device has been investigated to develop a drift compensation algorithm in order to handle the sensor drift. The calibration method uses drift compensation algorithms based on curve fitting of the temporal variation of the sensor signal of calibration samples. This procedure eliminates sensor drift within a single measurement sequence and over several sequences (days, months). It is also demonstrated that the algorithm preserves real features in the data structure of real samples that have been measured. However, this requires calibration samples that are highly correlated in sensor response with the real samples to be analysed in order to make a proper drift correction. ©2000 Elsevier Science B.V. All rights reserved.

**Keywords:** Gas-sensors; Drift; Calibration; Algorithms

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### 1. Introduction

Gas-sensor array technology combined with multi-variate data processing methods has a promising potential for rapid non-destructive analysis of vapours in a number of environments [1–8]. These systems should allow a high number of samples to be analysed within a short period of time with sufficient reproducibility and accuracy. As with all new techniques, there remain some basic problems to be solved to fulfill these requirements. One fundamental issue of the gas-sensor performance is the instability, better known as sensor drift, that leads to a poor reproducibility of the sensor signal [9,10]. This drift results in significant

temporal response variation of reference gas-samples, requiring frequent calibration of the system performance. Different reversible or irreversible processes may cause sensor drift. One example of a reversible process is the exposure of semiconductor based sensors to sulphur compounds or acids that may cause irreversible bindings on the sensor surface, leading to a change in sensor properties. Ageing is another irreversible alteration that may take place on the sensor material. This is a long-term deterioration caused by dismantling of the sensor surface material which distorts the sensitivity of the sensor. A reversible process that may influence the stability of sensor responses temporarily is the condensation of vapours on the sensor surface that may also affect the sensitivity. Once the measurement process is finished, the condensate vaporises from the sensor surface. If the condensate

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contains no sensor poisoning components, no permanent damage is done to the sensor surface.

So far, it has not been possible to fabricate chemical sensors without drift. Accordingly, sensor drift has to be treated in some way in order to achieve reliable measurement data from gas-sensor arrays. One way of handling the drift of chemical sensors is by mathematical processing of the sensor signals. Different methods have been suggested recently to compensate for sensor drift in experiments for gas identification. They have been based on either a univariate (one single sensor) or multivariate (multiple sensors) approach [11–18]. A univariate parametric drift compensation method based on the time-dependent drift in baseline resistance of metal-oxide gas-sensors has been described previously [18]. The method was used to compensate for the day-to-day baseline drift over a 12-day measurement experiment. This approach, however, will not be useful for compensating baseline drift within a single measurement sequence with a continuously increasing baseline signal. This is frequently the case in real applications when a great number of samples need to be measured in a short time. Another univariate method is sensor signal processing using auto-regressive moving average (ARMA) and identification of gas samples with adaptive methods [16,17]. Self-organising classifiers that are based on the adaptation of reference patterns is one of the multivariate methods [15]. However, the last two methods represent predictive approaches with model learning phases and error prediction for handling the sensor drift, with limited usefulness for practical industrial applications.

There is still a need for a simple method that can compensate for both short time drift throughout a measurement sequence and long-term drift (days, months) and that does not need time consuming and complex learning processes for a calibration procedure. The objective of this study was to establish a simple direct computed algorithm that allows a quick calibration of the measurement data to compensate for both short-term (hours) and long-term (days, months) sensor drift.

Two different drift counteraction methods have, therefore, been applied to correct sensor drift. One algorithm that corrects drift within a single measurement sequence, which is usually not much longer than one to a few hours, and a second algorithm that

corrects sensor drift over several sequences, which have been measured over days or months.

## 2. Material and methods

### 2.1. Gas-sensor array

Two commercial hybrid gas-sensor array instruments manufactured by Nordic Sensor Technologies A.B, Linköping have been used for this study; one system (NST 3220) where samples have been measured at ambient temperature and another system (NST 3320) where the samples have been heated prior to measurement. The sensor configuration of these systems consisted two blocks of five MOSFET sensors set at 140 and 170°C, respectively, one block with five Taguchi semiconductors set at 400°C coupled in series. The MOSFET (metal-oxide semiconductor field-effect transistor) gas-sensors consist of three layers; a doped silicon semiconductor, a thick oxide layer ( $\text{SiO}_2$ ) as insulator and on top a thin catalytic metal layer [19]. The catalytic metal consists of Pd, Ir or Pt and varies from 5 to 35 nm in thickness [7]. When gas molecules interact with the catalytic surface, a change in potential over the sensor due to electrical polarisation occurs. The output signal corresponds to the change in voltage required to keep a constant preset drain current. The sensitivity of MOSFET gas-sensors is determined by the operating temperature, metal layer thickness and the kind of metal used. The Taguchi sensors are metal-oxide semiconductor gas-sensors. They consist of a metal-oxide semiconducting film coated on to a ceramic substrate (alumina) with an integrated heater [20]. The gas sensing principle is based on the reaction between adsorbed ionised oxygen on the oxide surface with incoming oxidising or reducing molecules at elevated temperature. The interactions include reactions from chemisorption to catalytic combustion. The output signal is derived by a change in conductivity of the oxide caused by the reaction between the adsorbed oxygen and the incoming molecules.

### 2.2. Calibration samples

In principle, a sample should fulfill the following requirements in order to be applied as a calibration (reference) sample:

- It should be chemically stable over time, at least during a measurement sequence.
- It should be highly correlated with the real samples to be measured, i.e. it should respond to the same sensors and give signals within the same order of magnitude as the real samples.
- The sensor drift of calibration samples and real samples should be highly correlated.

### 2.3. Material and methods

The measurement data used in this study were taken from two different measurement experiments; the first data set was from a storage experiment on fish and the other data set from a measurement experiment on off-flavoured cow's milk.

The measured sensor response signals were used as test data for the drift compensation model.

### 2.4. Fish experiment

The fish storage experiment was carried out on fresh salmon samples in a temperature controlled room kept at  $4 \pm 0.5^\circ\text{C}$ , over 10 days. Data from the first 5 days of measurement have been used in this study. Distilled water samples and fish samples were kept in dark brown 250 ml glass vessels (Schott, Germany). 20 ml distilled water that had been equilibrated to 100% relative humidity was used as calibration sample that was replaced daily. The sample bottles were sealed with pre-heated ( $105^\circ\text{C}$ , 10 h) silicone septa and open screw caps. The measurement sequence consisted of five calibration samples measured after every sixth fish sample. One measurement sequence lasted for 1.5 h and was repeated every 24 h over 5 days.

### 2.5. Milk experiment

The milk data represented 2 days of measurements on two types of milk samples of different qualities. The objective was to discriminate fresh pasteurised milk from slightly oxidised pasteurised milk. The oxidised milk samples were prepared by bubbling synthetic air through pasteurised milk for 3–4 h with a flow of 800 ml/min at  $10^\circ\text{C}$ . 5 ml samples were kept in 30 ml glass vials and sealed with teflon coated silicone septa and open screw caps. One hundred and

ten milk samples were measured randomly over two measurement sequences over 2 days. Each sequence lasted for 2.5 h. Three pasteurised milk samples, from the start, middle and end of each sequence were used as calibration samples.

### 2.6. Gas measurements

The total measurement cycle for the fish experiment was 170:20 s reference air, baseline, 60 s sampling and 90 s recovery (reference air). The recovery included a 5 s flush of the sampling inlet system. The milk samples were incubated for 20 min at  $50^\circ\text{C}$  before gas measurement took place. The measurement conditions were 10 s reference air (baseline), 10 s sampling and 130 s recovery including 5 s flush.

Sample gas was transferred to the sensors by pumping the sample headspace gas with an auto-injector system into the sensor chambers. The flow rate of the membrane pump was set at 40 ml/min. Ambient air filtered with activated silica and charcoal was used as reference air during the baseline and recovery phases. The air filters were replaced daily.

### 2.7. Data analysis

The sensor response signal corresponding to the signal height relative to baseline was used as raw data for investigation of the sensor drift. Figures for visualising sensor drift were made by using the Excel programme (version 7.0). Gas-sensor response data were analysed statistically using The Unscrambler (version 6.11, CAMO A/S Trondheim, Norway): the line plots and principal component analysis (PCA) routines were used to study drift of single sensors and drift in the multidimensional sensor space. The Minitab for Windows (Version 12.1, Minitab) was used for statistical analysis of the sensor drift parameters.

## 3. Drift correction

### 3.1. Correction algorithms

The basic idea of drift compensation algorithms is to eliminate the variation in the sensor signal ascribed to sensor drift. This can be achieved by frequently

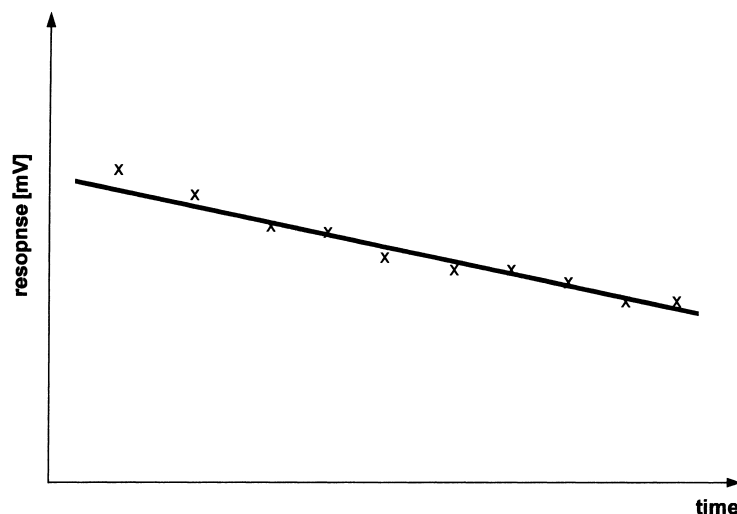


Fig. 1. Example of a typical temporal drift distribution of a sensor based on repeated measurements of a calibration sample.

measuring calibration samples. The temporal response variation of the calibration samples would describe the sensor drift that may be modelled by a mathematical time function. After drift correction, we should end up with identical sensor responses for the calibration samples, and also, drift in real sample responses should be eliminated.

The design of the suggested drift compensation method is based on the following criteria: the sensor drift follows an approximately linear drift course with time, each sensor has a unique drift characterisation and the drift should be modelled and compensated by means of a simple computational approach.

### 3.2. Sequence correction

Fig. 1 shows a typical temporal drift distribution for one sensor during repeated measurements of a calibration sample during a measurement sequence. The drift is characterised by a decrease in sensor response with time. In order to make corrections possible, calibration samples have to be measured in between the real samples throughout the measurement sequence. The line in Fig. 1 indicates the progress of the sensor drift during a sequence. A measurement sequence example with both real samples and calibration samples, that would allow sequence correction, is shown in Table 1.

The measurement starts with a calibration sample, followed by a certain number of real samples. Then, a new calibration sample is measured, followed again by several real samples. This continues until all real samples have been measured. Eventually, the sequence ends with a calibration sample, so that a complete sequence is given that allows a drift correction for the whole sequence. It should be noted that the number of real samples between the calibration samples should be kept constant, in order to achieve a satisfactory compensations algorithm. The general designation for responses in the sequence is  $U_{nk}$  for real sample and  $U_{cal,nk}$  for calibration sample, where index  $n$  indicates the sample positioning

Table 1  
Example of a sequence that makes a correction possible

| $n$ | Name                 | Sensor 1      | Sensor 2      | Sensor 3      | Sensor $_k$   |
|-----|----------------------|---------------|---------------|---------------|---------------|
| 1   | calibration sample 1 | $U_{cal,11}$  | $U_{cal,12}$  | $U_{cal,13}$  | $U_{cal,1k}$  |
| 2   | real sample 1        | $U_{21}$      | $U_{22}$      | $U_{23}$      | $U_{2k}$      |
| 3   | real sample 2        | $U_{31}$      | $U_{32}$      | $U_{33}$      | $U_{3k}$      |
| 4   | real sample 3        | $U_{41}$      | $U_{42}$      | $U_{43}$      | $U_{4k}$      |
| 5   | real sample 4        | $U_{51}$      | $U_{52}$      | $U_{53}$      | $U_{5k}$      |
| 6   | calibration sample 2 | $U_{cal,61}$  | $U_{cal,62}$  | $U_{cal,63}$  | $U_{cal,6k}$  |
| 7   | real sample 5        | $U_{71}$      | $U_{72}$      | $U_{73}$      | $U_{7k}$      |
| 8   | real sample 6        | $U_{81}$      | $U_{82}$      | $U_{83}$      | $U_{8k}$      |
| 9   | real sample 7        | $U_{91}$      | $U_{92}$      | $U_{93}$      | $U_{9k}$      |
| 10  | real sample 8        | $U_{101}$     | $U_{102}$     | $U_{103}$     | $U_{10k}$     |
| 11  | calibration sample 3 | $U_{cal,111}$ | $U_{cal,112}$ | $U_{cal,113}$ | $U_{cal,11k}$ |

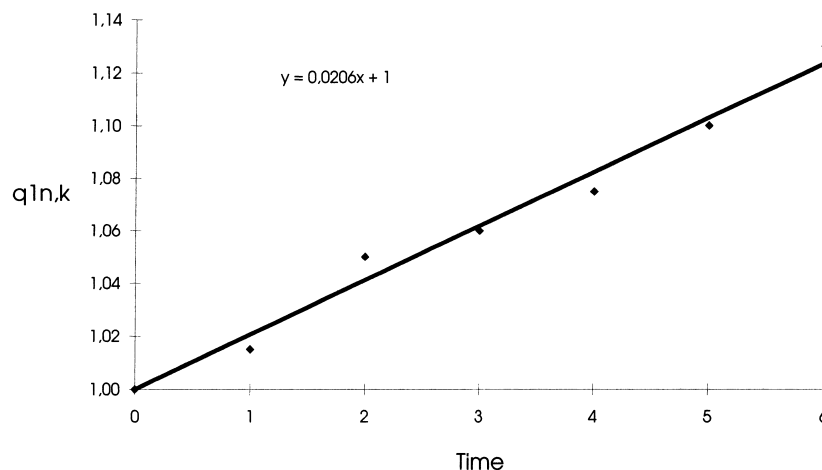


Fig. 2. Trendline fitted drift quotients  $q_{1n,k}$ , for sensor  $k$  with drift equation.

in the sequence and index  $k$  indicates the particular sensor.

Every individual sensor has a unique drift course and it is therefore required that a time dependent drift curve is fitted for every sensor selected for any application. In Eq. (1), the response  $U_{cal,1k}$  of the first calibration sample is divided by the response  $U_{cal,nk}$  of the second calibration sample in the measurement sequence ( $U_{cal,6k}$ , Table 1). The division would give a quotient  $q_{16,k}$  of the two first calibration responses for a particular sensor  $k$ . Next, the response  $U_{cal,1k}$  of the first calibration sample is divided by response  $U_{cal,nk}$  of the third calibration sample ( $U_{cal,11k}$ , Table 1), which gives another quotient  $q_{1n,k}$  ( $q_{111,k}$ , Table 1) for sensor  $k$ . By using the inverse of the quotients, this would lead to biased corrected sensor signals since we would have to divide by the predicted correction factor instead of multiplying with it. Continuing this operation for all calibration samples gives a number of quotients, one less than the number of calibration samples listed in the sequence:

$$q_{1n,k} = \frac{U_{cal,1k}}{U_{cal,nk}} \quad (1)$$

This procedure is to be carried through for all  $k$  sensors, giving  $k$  sensor drift diagrams, where each diagram shows the drift quotients  $q_{1n,k}$  for a single sensor. One example is given in Fig. 2, where the quotients are arranged by order of time. The diagram

starts with quotient  $q_{11,k}$ , calculated by dividing the first calibration sample response in the sequence for sensor  $k$  with itself, which is the starting point with value 1.

By linear regression of the quotients as a function of time, a trendline is fitted to the data points that corresponds best with the calculated quotients  $q_{1n,k}$  and with intercept 1.0. The equation of the trendline that describes the drift for a particular sensor  $k$  can be used to correct all measured sample responses in the sequence for this sensor. The power of the equation should not exceed the value 3. Higher values would lead to more complex curves that do not necessarily describe the trend better, but rather contribute to noise.

Eq. (2) represents the trendline equation determined in the quotient diagram, assuming a linear function. As mentioned, the power of the trendline equation could also be higher, but for most cases, a linear approach seems to be valid provided that the responses are within the linear dynamic response range of the sensors. Eq. (2) gives the correction factor  $f_{nk}$  of sample  $n$  for sensor  $k$ , that will be used later in Eqs. (3a), (3b), (3c) and (3d):

$$f_{nk} = (ax_n + b)_k \quad (2)$$

where  $f_{nk}$  is the correction factor for the response of calibration sample or real sample at position  $n$  in the measurement sequence for sensor  $k$ ,  $x_n$  the time value for calibration sample or real sample at position  $n$  in

Table 2  
Example of  $x_n$ -values for calibration samples

| Name                 | $x_n$ |
|----------------------|-------|
| Calibration sample 1 | 0     |
| Real sample 1        |       |
| Real sample 2        |       |
| Real sample 3        |       |
| Real sample 4        |       |
| Calibration sample 2 | 1     |
| Real sample 5        |       |
| Real sample 6        |       |
| Real sample 7        |       |
| Real sample 8        |       |
| Calibration sample 3 | 2     |

measurement sequence,  $a$  the trendline gradient and  $b$  the trendline intercept. Since the  $x$ -axis shows the quotients  $q_{1n,k}$  in order of time, all the quotients correspond to a time value  $x_n$  in Eq. (2). Measuring the calibration sample regularly implies that there is always the same number of real samples between the calibration samples in the sequence in order to describe the drift correctly. Constant time between the calibration samples is not essential for accurate drift modelling as long as a high enough frequency of calibration samples is included throughout the whole sequence. In our case, just for simplicity reasons, the calibration samples always have whole numbers for  $x_n$ . The first calibration sample has value 0 for  $x_n$ , the second calibration sample has value 1 for  $x_n$ , and so on as displayed in Table 2.

All real samples between two calibration samples have  $x_n$ -values between the  $x_n$ -values of the two calibration samples. When the second calibration sample, for example, has  $x_n$ -value 1 and the third calibration sample has  $x_n$ -value 2, all the real samples between them would have  $x_n$ -values between 1 and 2. Thus, for all real samples, the  $x_n$ -value can be interpolated. Mathematically, this requires identical cycle time for both calibration sample and real sample. Table 3 shows such a value distribution for  $x_n$ .

In Eq. (3a) or Eq. (3b), all measured real sample responses  $U_{nk}$  and in Eq. (3c) or Eq. (3d), all calibration sample responses  $U_{cal,nk}$  of sensor  $k$  now become multiplied with their specific correction factors  $f_{nk}$  that Eq. (2) for this sensor gives. The results are new values  $U_{seq,corr,nk}$  and  $U_{seq,corr,cal,nk}$ , respectively, for the responses, where the drift will not be expressed

Table 3  
Example of  $x_n$ -values for real samples between two calibration samples

| Name                 | $x_n$ |
|----------------------|-------|
| Calibration sample 2 | 1     |
| Real sample 5        | 1,2   |
| Real sample 6        | 1,4   |
| Real sample 7        | 1,6   |
| Real sample 8        | 1,8   |
| Calibration sample 3 | 2     |

any more within one measurement sequence:

$$U_{seq,corr,nk} = U_{nk} f_{nk} \quad (3a)$$

$$U_{seq,corr,nk} = U_{nk} (ax_n + b)_k \quad (3b)$$

$$U_{seq,corr,cal,nk} = U_{cal,nk} f_{nk} \quad (3c)$$

$$U_{seq,corr,cal,nk} = U_{cal,nk} (ax_n + b)_k \quad (3d)$$

where  $U_{seq,corr,nk}$  is the sequence corrected response of real sample at position  $n$  in measurement sequence for sensor  $k$ .  $U_{nk}$  the response of real sample at position  $n$  in measurement sequence for sensor  $k$ .  $U_{seq,corr,cal,nk}$  is the sequence corrected response of calibration sample at position  $n$  in measurement sequence for sensor  $k$ , and  $U_{cal,nk}$  the response of calibration sample at position  $n$  in measurement sequence for sensor  $k$ .

Fig. 3 displays graphically how the responses of the calibration samples later in the sequence get aligned to the response level of the first calibration sample (indicated by arrows). The thick line shows the drift of repeated calibration samples, as in Fig. 1. The responses of the real samples measured in between the calibration samples also get aligned, but they are not shown in Fig. 3 since their responses will differ.

### 3.3. Multisequence correction

Plotting the response of repeated calibration samples for one sensor of several sequences could look as in Fig. 4 (bold lines). The responses are decreasing in all sequences, but they are at different levels. Before the multisequence correction can be done, the sequence correction for each sequence has to be carried out. That is, aligning the responses of calibration samples within one sequence to the same level and

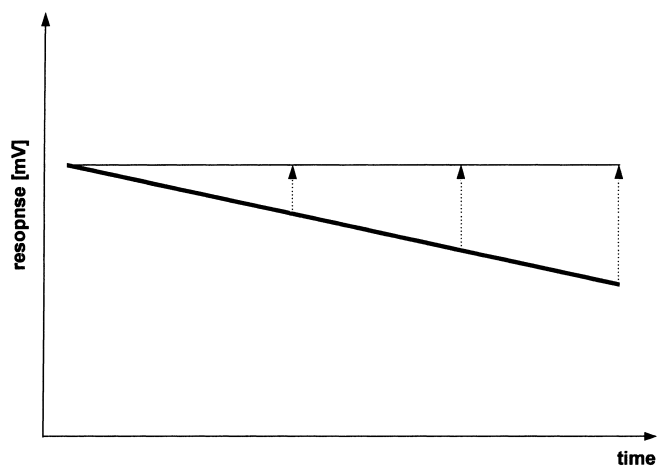


Fig. 3. Elimination of drift by multiplying responses with a correction factor  $f_{nk}$ . Arrows indicate alignment of the sensor signal after drift correction.

correcting all real samples in the same process, as described earlier.

After correction within the individual sequences, all the calibration sample responses within each sequence will be on a level which would still differ between the sequences. Further, all the levels need to be aligned on the same level as the first sequence. This is done in Eq. (5a) or Eq. (5c) for the sequence corrected real sample responses  $U_{\text{seq,corr},nk}$  and in Eq. (5b) or Eq. (5d) for the sequence corrected calibration sample responses  $U_{\text{seq,corr,cal},nk}$  by multiplying with another correction

factor  $f_{ks}$ . The correction factor  $f_{ks}$  for a particular sequence is the quotient of the first calibration sample response  $(U_{\text{cal},1k})_1$  in the first sequence divided by the first calibration sample response  $(U_{\text{cal},1k})_s$  in sequence  $s$  which is to be corrected as displayed in Eq. (4):

$$f_{ks} = \frac{(U_{\text{cal},1k})_1}{(U_{\text{cal},1k})_s} \quad (4)$$

where  $f_{ks}$  is the multiplying factor for all responses of sensor  $k$  in sequence  $s$ .

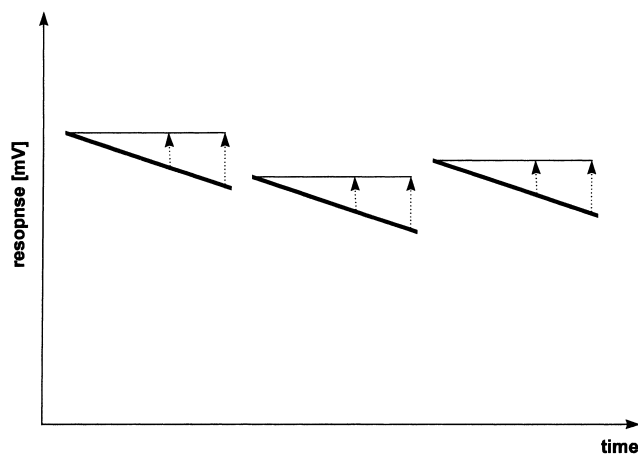


Fig. 4. Sequence correction for three measurement sequences. Arrows indicate alignment of the sensor signal within each sequence after drift correction.

The result is that all calibration sample responses will be aligned on the same level and all real samples in between can be considered as drift corrected:

$$(U_{\text{corr},nk})_s = (U_{\text{seq,corr},nk})_s f_{ks} \quad (5a)$$

$$(U_{\text{corr,cal},nk})_s = (U_{\text{seq,corr,cal},nk})_s f_{ks} \quad (5b)$$

$$(U_{\text{corr},nk})_s = [U_{nk}(ax_n + b)_k]_s \frac{(U_{\text{cal},1k})_1}{(U_{\text{cal},1k})_s} \quad (5c)$$

$$(U_{\text{corr,cal},nk})_s = [U_{\text{cal},nk}(ax_n + b)_k]_s \frac{(U_{\text{cal},1k})_1}{(U_{\text{cal},1k})_s} \quad (5d)$$

where  $(U_{\text{corr},nk})_s$  is the final corrected response of real sample at position  $n$  in measurement sequence  $s$  for sensor  $k$  and  $(U_{\text{corr,cal},nk})_s$  the final corrected response of calibration sample at position  $n$  in measurement sequence  $s$  for sensor  $k$ .

## 4. Results and discussion

### 4.1. Selection of sensors for drift correction

Table 4 lists the responses of the calibration samples from 6 February and the respective quotients  $q_{1n,k}$ . In this experiment, the Taguchi sensors were set at a low scaling factor explaining the relatively low water responses for these sensors. The quotient values have been derived by Eq. (1). In addition, drift equations have been determined for each sensor of the array (Table 5) based on linear regression of the quotient values given in Table 4. The following criteria have been examined for compensation of sensor drift (Table 6): the magnitude of the sensor response, the variation of the drift quotient ( $\Delta q_{1n,k}$ ) and the statistical significance of the slope of the drift equation. These three quantities are decisive when it comes to selecting sensors for drift correction. A low  $S/N$  ratio will imply that a significant proportion of the drift is influenced by noise and this may add redundant information to the sensor signal when it is being corrected. A sensor with a high quotient variation ( $\Delta q_{1n,k}$ ), reflecting a relatively high variation in response, could lead to an overcompensation of the sensor signal. A sensor signal that does not show any systematic temporal drift would not require any drift correction since the temporal variation of the signal would be random, i.e. the

slope of the drift equation would not be statistically significant. We have applied the following quantitative criteria that must be fulfilled to qualify for drift correction: the signal to noise ratio of a sensor response has to be greater than 3. The noise has been defined as the mean background response plus one standard deviation. The variation of the drift quotient should be less than 0.100 and the slope of the drift equation should be statistically significant within a 95% confidence level ( $p < 0.05$ ).

From the inspection of Table 4, it is seen that, of the whole sensor array, only MOSFETs 2–3 and 5–8 and TGS 1 fulfill all the three selection criteria. They have responses varying from 15 to 158 in signal to noise ratio, the  $\Delta q_{1n,k}$  values are less than 0.1 varying from 0.027 to 0.081 and the statistical significance of the slope of the drift equations vary between 0.001 and 0.033. For the MOSFETs 1, 4 and 9 and TGSs 2, 4 and 5, none of the three selection criteria are fulfilled and they will therefore be excluded from drift correction.

MOSFET 4 is an extreme example for the latter type of sensor responses. The drift quotients vary in a range from 0.526 to 3.699 which is about 60 times more than the average variation of 0.057 for the first group of sensors that fulfilled the selection criteria. In addition, it is strongly influenced by noise ( $S/N = 1.5$ ) and the slope of the drift equation is not statistically significant. Drift correction for this sensor would obviously introduce an overcompensation of the sensor signal and introduce redundant information in drift corrected measurement data.

MOSFET 10 is a borderline case with a reasonable response ( $S/N = 12.8$ ), a high  $\Delta q_{1n,k}$  exceeding 0.100 and a slope that is statistically significant ( $p = 0.002$ ). Despite the high  $\Delta q_{1n,k}$  (0.547), we decided to select this sensor for correction. This is based on the fact that the quotients for this sensor increase continuously and thus explain a strong drift in the sensor signal, which also decreases continuously with time.

### 4.2. Calibration measurements

Provided that the drift is linear, a drift correction based only on a calibration measurement at the start and at the end of a measurement sequence would be sufficient in order to compensate for sensor drift. However, this would rarely be the case as we and others



Table 4

Calibration samples of the fish experiment from 6 February (first day of measurement in the experiment) with sensor responses and quotients  $q_{1n,k}$ <sup>a</sup>

| Name        | MOSFET 1 | MOSFET 2 | MOSFET 3 | MOSFET 4 | MOSFET 5 | MOSFET 6 | MOSFET 7 | MOSFET 8 | MOSFET 9 | MOSFET 10 | TGS 1 | TGS 2 | TGS 3 | TGS 4 | TGS 5 |
|-------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|-----------|-------|-------|-------|-------|-------|
| 6c1         | 2.87     | 12.91    | 24.93    | 0.56     | 12.40    | 15.57    | 9.12     | 29.80    | 1.16     | 3.80      | 0.04  | 0.28  | 1.26  | 0.03  | 0.16  |
| 6c2         | 2.69     | 12.59    | 24.77    | 0.33     | 12.54    | 15.47    | 8.84     | 29.59    | 1.14     | 3.44      | 0.04  | 0.30  | 1.30  | 0.03  | 0.17  |
| 6c3         | 2.63     | 12.33    | 23.67    | 0.28     | 11.63    | 15.21    | 8.97     | 28.73    | 1.04     | 3.22      | 0.04  | 0.27  | 1.17  | 0.03  | 0.15  |
| 6c4         | 3.32     | 12.28    | 23.36    | 1.07     | 11.58    | 15.52    | 8.71     | 28.57    | 1.12     | 3.12      | 0.04  | 0.28  | 1.18  | 0.03  | 0.16  |
| 6c5         | 2.54     | 12.23    | 23.30    | 0.15     | 11.62    | 15.17    | 8.86     | 28.50    | 1.11     | 2.46      | 0.04  | 0.28  | 1.19  | 0.03  | 0.16  |
| $q_{11,k}$  | 1.000    | 1.000    | 1.000    | 1.000    | 1.000    | 1.000    | 1.000    | 1.000    | 1.000    | 1.000     | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| $q_{18,k}$  | 1.068    | 1.025    | 1.007    | 1.690    | 0.989    | 1.006    | 1.032    | 1.007    | 1.025    | 1.106     | 0.994 | 0.931 | 0.968 | 0.989 | 0.965 |
| $q_{115,k}$ | 1.093    | 1.047    | 1.054    | 2.019    | 1.066    | 1.024    | 1.017    | 1.037    | 1.117    | 1.180     | 1.068 | 1.052 | 1.074 | 1.108 | 1.080 |
| $q_{122,k}$ | 0.865    | 1.051    | 1.067    | 0.526    | 1.070    | 1.003    | 1.047    | 1.043    | 1.042    | 1.219     | 1.033 | 1.024 | 1.061 | 0.981 | 1.022 |
| $q_{129,k}$ | 1.131    | 1.056    | 1.070    | 3.699    | 1.067    | 1.027    | 1.029    | 1.045    | 1.049    | 1.547     | 1.045 | 0.999 | 1.055 | 1.059 | 1.012 |

<sup>a</sup>The first number of the name is the day of measurement (6 February) followed by the letter c for calibration and a second number corresponding to the order (1–5) within the sequence.

Table 5

Results from regression analysis of correction coefficients ( $q_{1n,k}$ ) for all the sensors from the first measurement sequence. Drift equation, correlation coefficient ( $r$ ), standard deviation of slope and observations,  $p$ -value and number of measurements ( $n$ ) are listed

| Sensor    | Drift equation    | $r$  | s.d. slope | s.d. observed | $p$ -value | $n$ |
|-----------|-------------------|------|------------|---------------|------------|-----|
| MOSFET 1  | $y = 0.0124x + 1$ | 0.09 | 0.0195     | 0.1049        | 0.552      | 5   |
| MOSFET 2  | $y = 0.0165x + 1$ | 0.90 | 0.0017     | 0.0096        | 0.001      | 5   |
| MOSFET 3  | $y = 0.0198x + 1$ | 0.94 | 0.0021     | 0.0113        | 0.001      | 5   |
| MOSFET 4  | $y = 0.40340 + 1$ | 0.55 | 0.1858     | 1.0180        | 0.096      | 5   |
| MOSFET 5  | $y = 0.0199x + 1$ | 0.84 | 0.0040     | 0.0219        | 0.007      | 5   |
| MOSFET 6  | $y = 0.0056x + 1$ | 0.63 | 0.0017     | 0.0095        | 0.033      | 5   |
| MOSFET 7  | $y = 0.0108x + 1$ | 0.55 | 0.0027     | 0.0149        | 0.017      | 5   |
| MOSFET 8  | $y = 0.0131x + 1$ | 0.94 | 0.0013     | 0.0072        | 0.001      | 5   |
| MOSFET 9  | $y = 0.0194x + 1$ | 0.42 | 0.0078     | 0.0425        | 0.067      | 5   |
| MOSFET 10 | $y = 0.1103x + 1$ | 0.92 | 0.0145     | 0.0794        | 0.002      | 5   |
| TGS 1     | $y = 0.0136x + 1$ | 0.66 | 0.0043     | 0.0234        | 0.033      | 5   |
| TGS 2     | $y = 0.0034x + 1$ | 0.32 | 0.0080     | 0.0438        | 0.690      | 5   |
| TGS 3     | $y = 0.0173x + 1$ | 0.77 | 0.0060     | 0.0327        | 0.044      | 5   |
| TGS 4     | $y = 0.0128x + 1$ | 0.32 | 0.0094     | 0.0517        | 0.247      | 5   |
| TGS 5     | $y = 0.0079x + 1$ | 0.31 | 0.0073     | 0.0398        | 0.335      | 5   |

[13,14] have experienced with solid-state based sensors. There may occur a significant non-linear drift during the first measurements of a sequence before the sensors get stabilised at an approximately linear drifting course. In our case, the sensors were always exposed to three to five calibration samples before a measurement sequence was started. This was sufficient in order to get the sensors adjusted to a linear drifting course. It is therefore essential that real samples are measured and drift compensation is performed after the adjustment measurements in order to avoid an overcompensation of the sensor drift.

Fig. 5 shows the response course of calibration samples over five days (6–10 February) for MOSFET 3, which was similar for all other sensors in the sensor array that gave significant response for the calibration samples. The day-to-day variation of the responses caused by sensor drift clearly demonstrates the need for drift correction in order to achieve results that are sensible and that allow for revealing features in real sample data.

The equations of the temporal sensor drift for the selected sensors on 6 February can be seen in Table 5. After correction, the response course of MOSFET 3 for the calibration samples looked as in Fig. 6. The drift range of the response varied between 20 and 27 mV, a total of 7 mV. Still, the structure of the response pattern of the samples within each day is basically remaining when compared to those in Fig.

5 for the uncorrected responses. The remaining variation reflects the instrumental precision of replicate measurements within a sequence which in this case is within 5% of the absolute sensor response value of the corrected samples.

PCA has been carried out on the uncorrected and corrected sensor responses of calibration samples including the sensors selected for drift correction. PCA score plots of the uncorrected and corrected data are shown in Figs. 7 and 8. The calibration measurements should lie close to each other with no sign of any day-to-day clustering in the PCA score plot which is obviously not the case in the uncorrected data. A clear day-to-day clustering of the calibration samples can be seen in the uncorrected data (Fig. 7). The day-to-day drift of the sensor signal that was seen in the single sensors is also kept in the PCA score plot based on the selected sensors. The clusters from 8–9 February are overlapping, which is also in agreement with what was observed for a single sensor (Fig. 5). After drift correction by applying the drift equations on the selected sensors, the result as displayed in the PCA plot (Fig. 8) is achieved. No more day-to-day clustering of samples occurs. Another proof that the correction algorithm fulfills the demands in eliminating the sensor drift is demonstrated by the decreasing explained total variance of PC 1 and PC 2 after drift correction. Before correction, PC 1 and PC 2 explained 94% (84% by PC 1 and 10% by PC 2) of the variance in the response

Table 6  
List of the numeric values of drift correction selection criteria for each sensor of the array<sup>a</sup>

| Name              | MOSFET 1 | MOSFET 2 | MOSFET 3 | MOSFET 4 | MOSFET 5 | MOSFET 6 | MOSFET 7 | MOSFET 8 | MOSFET 9 | MOSFET 10 | TGS 1 | TGS 2 | TGS 3 | TGS 4 | TGS 5 |
|-------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|-----------|-------|-------|-------|-------|-------|
| $S/N$             | 14.2     | 31.7     | 56.8     | 1.5      | 29.8     | 34.7     | 15.0     | 87.6     | 2.5      | 12.8      | 23.2  | 13.0  | 158.0 | 26.4  | 14.6  |
| $\Delta q_{1n,k}$ | 0.266    | 0.056    | 0.070    | 3.173    | 0.081    | 0.027    | 0.047    | 0.045    | 0.117    | 0.547     | 0.074 | 0.121 | 0.106 | 0.127 | 0.115 |
| $p$               | 0.552    | 0.001    | 0.001    | 0.096    | 0.007    | 0.033    | 0.017    | 0.001    | 0.067    | 0.002     | 0.033 | 0.690 | 0.044 | 0.247 | 0.335 |

<sup>a</sup>  $S/N$  represents the signal to noise ratio of the mean response from Table 4.  $\Delta q_{1n,k}$  is the quotient variation range of a sensor based on quotient data in Table 4 and  $p$  is the statistical significance of the slope of the drift equation from Table 5.

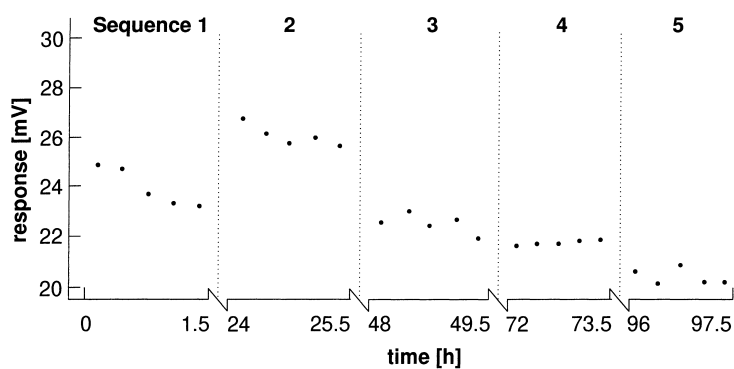


Fig. 5. Lineplot with drifting response signals of calibration samples for MOSFET 3 during 5 days of measurement (sequence 1–5).

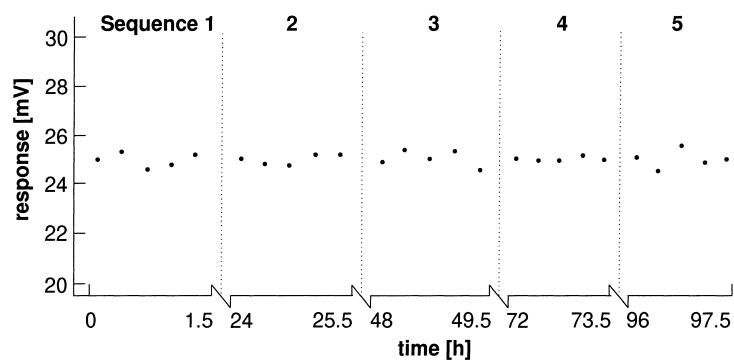


Fig. 6. Lineplot with drift corrected response signals of calibration samples for MOSFET 3 (Fig. 5) during 5 days of measurement (sequence 1–5).

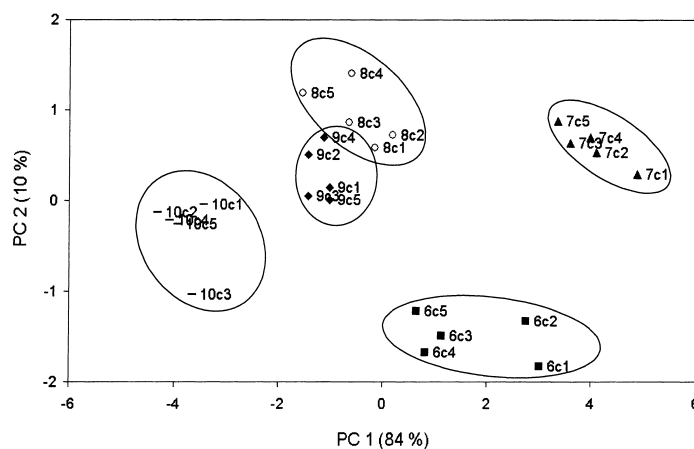


Fig. 7. PCA score plot of uncorrected calibration samples measured daily over 5 days. The first number of the sample label represents the day of measurement (6–10 February) followed by the letter c for calibration and a second number corresponding to the order (1–5) of measurement within the sequence.

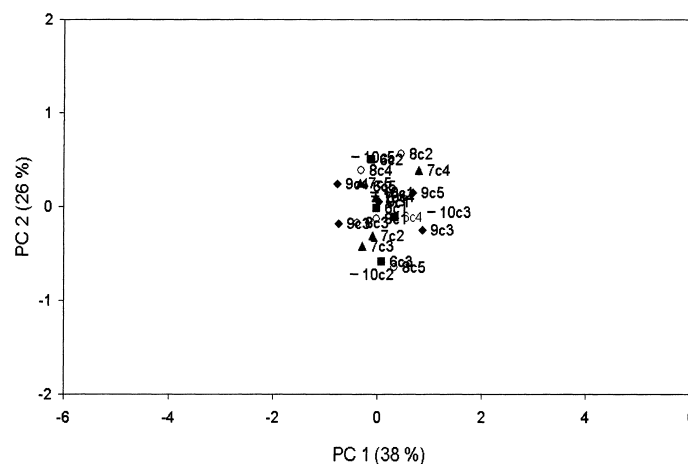


Fig. 8. PCA plot of corrected calibration samples measured daily over 5 days. The first number of the sample label is the day of measurement (6–10 February) followed by the letter c for calibration and a second number corresponding to the order (1–5) of measurement within a sequence.

data (Fig. 7). After correction, however (Fig. 8), the total explained variance by PC 1 and PC 2 is a mere 64% (38% by PC 1 and 26% by PC 2). The greatest variation, the drift, is taken out of the corrected data, and thus, the two greatest of the remaining variations do not explain as much variation of the whole data set as prior to correction. Accordingly, a significant decrease in total variance explained by PC 1 and PC 2 and a great decrease in the explained variance by PC 1 alone after the correction are additional evidence that sensor drift is eliminated for the calibration samples.

#### 4.3. Fish measurements

Fig. 9 shows a line plot of the response of uncorrected fish samples from 6 February to 10 February (sequence 1–5), i.e. the same time period as the calibration samples (Fig. 5) for MOSFET 3. The responses for the different fish samples are relatively constant for 6 February. For 7 February, the responses are at a higher level, but still fairly constant, and back at a lower level from 8 February. 10 February onwards, some responses increased significantly due to increased release of volatiles derived from bacterial decay of fish tissue.

In general, a day-to-day variation of the sensor responses over the first 3 days of measurement can be seen (Fig. 9) similar to the distribution of the calibra-

tion samples (Fig. 5) during the first 3 days of measurement until the increasing signals of the following days exceed the sensor drift seen for the calibration samples. Also, a slight decrease within a single sequence may be seen. Fig. 10 shows the responses for the same samples over the same period after drift correction. The responses of 7 February are now on the same level as those of the day before and after and there is no abrupt day-to-day break in the line of responses as was observed for the uncorrected data in Fig. 9. It is also seen that, after correction, the response signal of the fish samples increases slightly throughout the whole measurement period as would be expected during storage due to increased release of volatile compounds into the sample headspace. A similar time trend could not be detected in the uncorrected data since they were strongly influenced by sensor drift masking the systematic time variation in the data. Also, as for the calibration samples, it is seen that the general structure of the response pattern is maintained when compared to the uncorrected data (Fig. 9), i.e. that information of real sample variation is not lost after drift correction.

#### 4.4. Milk measurements

Two of the Taguchi sensors TGS that showed the greatest discrimination ability were sufficient in order to discriminate between the reference (pasteurised)

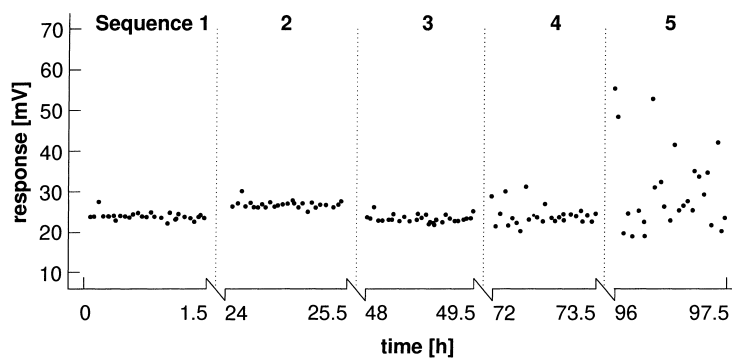


Fig. 9. Lineplot of uncorrected fish sample responses for MOSFET 3 measured over 5 days (sequence 1–5).

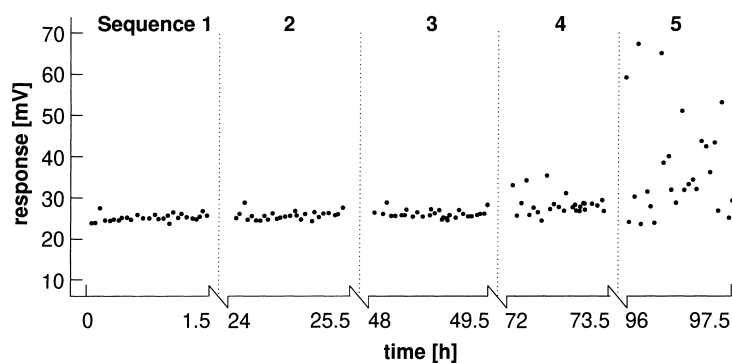


Fig. 10. Lineplot of corrected fish sample response signals for MOSFET 3 measured over 5 days (sequence 1–5).

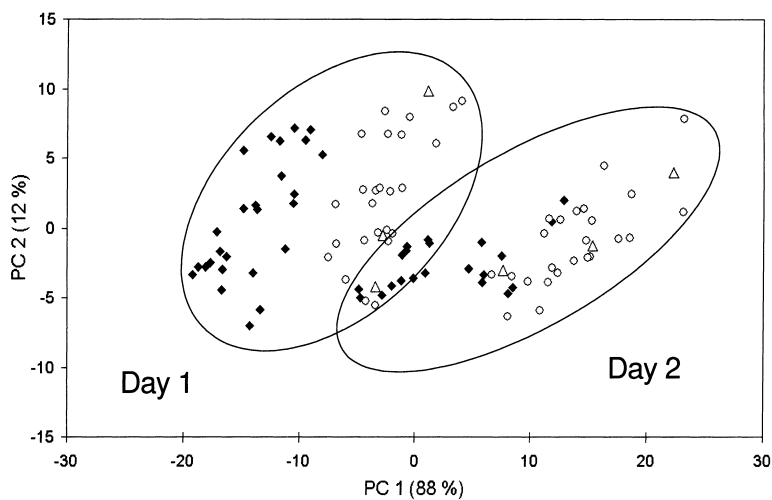


Fig. 11. PCA plot of uncorrected milk samples measured over 2 days; ○: pasteurised milk, ◆: oxidised pasteurised milk, △: calibration samples.

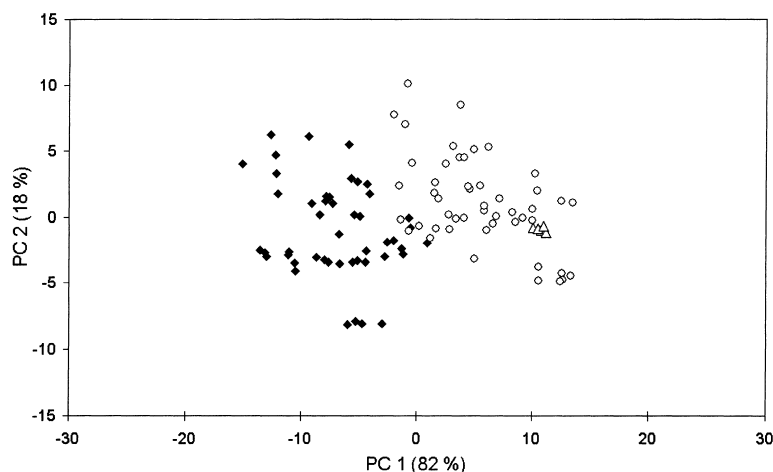


Fig. 12. PCA plot of drift corrected milk samples measured over 2 days; ○: reference milk, ◆: oxidised milk, △: calibration samples.

and oxidised milk samples. A PCA plot of the measurement data over the 2 days before and after calibration are shown in Figs. 11 and 12. A typical day-to-day clustering is seen between the two sequences due to sensor drift. The measurements from the first day show a good discrimination despite sensor drift that to a great extent is described along the second component. The oxidised samples tend to be localised to the left-hand side of the PCA plot along axis 1. The data from the second day show a slightly poorer discrimination compared to the first day, with a few overlapping samples in both quality classes. This may partly be ascribed to sensor drift that tends to be expressed more along the first axis than was observed for the first day. After applying the drift correction algorithm based on the multisequence correction algorithm to these two sensors, the result as shown in Fig. 12 was obtained. The samples are now clustered in two groups according to their quality along the first principal component. The oxidised samples become grouped to the left-hand side and the reference samples grouped to the right-hand side along the first axis. A few samples still seem to overlap after drift correction, suggesting that no clear discrimination could be obtained for these few samples.

#### 4.5. Loss of information through sensor selection

During the first 4 days of the fish measurements, the fish samples give responses on the same sensors as

the calibration samples as shown for MOSFET 3 (Fig. 9) and in Table 4. From the fifth day of measurement, when the bacterial spoilage of the fish becomes more apparent, sensors start giving significant responses for some of the fish samples. Since the calibration samples allow correction just for the responses of MOSFETs 2–3, 5–8 and 10, only the fish sample responses of the same sensors can be applied for correction. The responses of the remaining sensors can not be corrected and thus have to be kept out of the correction, even if they deliver high values for the fish samples that may contain a systematic variation with time. This may imply a loss of information which may affect the PCA model and the distribution of the samples in the PCA score plot.

This makes it very clear that the calibration sample should in principle be highly correlated with the real sample. Loss of information can be avoided since the calibration sample and real sample would have responses for the same sensors and correction for all these responses could be done. This was the case for the selected sensors used for correction (Fig. 5). The ideal case, however, would be to have a stable calibration sample that gives response over the entire sensor array similar to the samples being measured. This can be achieved by preparing calibration samples made of gas mixtures or real samples that are stable over time and that give a similar sensor response pattern as the real samples to be analysed.

#### 4.6. Baseline distortion

Another limitation that has to be mentioned concerning drift correction of sensor response signals is the distortion of the sensor baseline that may occur when sensors are being exposed to highly concentrated vapours. As long as the fish samples did not give very high responses, the calibration samples showed constant responses, entirely influenced by sensor drift. From the measurement on the seventh day of storage, the calibration sample responses could not be considered stable any longer because their baseline values were much higher than expected and their response values varied far more than during the first 6 days of the experiment (data not shown). This effect of sensor baseline distortion may be caused by several factors. One is a very slow baseline recovery triggered by high molecular weight compounds that do not leave the sensor surface during the recovery phase [21]. The recovery time for the sensors used during the experiment seemed to be too short to re-establish proper baseline conditions during the later phase of the experiment and the baseline signal stayed at far higher values than normal. Another more likely explanation could be a memory effect of the sample inlet system due to cross contamination of calibration samples by a previously measured highly concentrated fish sample.

This kind of distortion may have affected the response signal of the calibration samples during the late stage of the experiment and delivered wrong response values for the calibration samples which made them useless for drift correction. To avoid baseline distortions caused by shorter recovery time, optimising sample amount and conditioning and longer flushing of the inlet system would be necessary in order to keep the responses within a reasonable response range, where such distortions do not occur. This should be done in a test sequence before long-time series of real samples are being handled in order to keep the samples within the linear dynamic range of the sensors.

#### 5. Conclusions

The present sensor technology provides gas-sensors that perform in a way that frequently makes post treat-

ment of measurement results necessary. Instability of response signals and ageing cause sensor drift that complicates long-term routine analysis with the usage of gas-sensor arrays. Several time consuming methods that require a lot of preparatory work to compensate for sensor drift have been developed already. This leaves the need for a quick data processing performed by a relatively simple algorithm to make an immediate investigation of the real samples possible.

The algorithm described in this study fulfills the requirements for a calibration routine that eliminates sensor drift: robustness, rapidity, simplicity and consistency in data feature. This has been demonstrated under real experimental conditions by regularly measuring calibration samples in between real samples throughout measurement sequences over several days. By simple modifications, this method can easily be applied to real time compensation of sensor drift. This could be done by periodically performing an automated updating of the drift quotients for every new calibration measurement combined with a test on the selection criteria. This would allow real-time calibration during measurement acquisition and be useful in on-line applications.

We have applied the signal height of the sensor transient to describe temporal sensor drift in this experiment. The same method could in principle also be applied to other transient parameters of the sensor signal [22], which we have experienced as being affected by temporal drift in the same way as the response signal.

In both experiments, a drift correction of the measurement data was required since the magnitude of the sensor drift masked the systematic variation in the data. The correction algorithm eliminated the sensor drift successfully, which was evident by response lineplots and PCA score plots. Since the correction could not be realised for all the sensors, this implied that some sensors had to be excluded from correction, which could lead to a loss in information. Finding a more suitable calibration sample that gives responses for all sensors of the sensor array similar to the real samples that have to be analysed is of utter importance to allow drift correction without information loss. However, in the case of the fish data, it was seen that, in the later stage of the storage, drift correction was not required, when the signals by far exceeded the drift of the sensors. Accordingly, depending on the resolution and the precision that are required in order



to elucidate systematic variation in the real samples, the extent to which a drift correction is required should always be considered.

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